Seeing Crypto Markets with Public LLMs: An Observation Layer that Compiles World Bundles

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Abstract

Purpose. Public LLMs fail on crypto before strategy: they cannot *see* a fragmented, asynchronous market. We contribute a crypto *observation layer*—an anonymized world-model runtime (state compiler + quality gate, not a generative simulator)—that turns multi-band evidence into a point-in-time (PIT) *world bundle* public models can consume. Hard refusal is only the safety valve: when support is thin, a machine-readable flag blocks action so half-baked context cannot drive decisions. **Evidence.** Primary: on identical OOS days, grounding recovers ready/missing/tilt at ≈0.97 under Compiled vs. ≲0.23 under Raw—the bundle makes the market checkably legible, not merely longer context. Secondary: across four live vendors (temperature 0), Compiled thin-abstain is 1.0 while Ungated soft judgment on the *same* content is only ≈0.68; Raw overtrades and loses; Blind refuses with no feed. The panel is scarce (max WMI ≈ 0.093<0.2), so full Compiled refusal is a valve stress test. A no-LLM rule on vintaged tilts beats momentum (ΔCE=0.334, p=0.034). **Claim.** Compile so models can see; refuse when they must not act.

CCS Concepts

• **Computing methodologies → Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics.*

Keywords

observation layer, world bundle, grounding, public LLMs, cryptocurrency, point-in-time, quality-gated refusal, systems

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1 Introduction

Purpose (read this first). Make public LLMs *see* crypto markets. Exchange microstructure, macro vintages, and alternative flows arrive on incompatible clocks [11, 13], while generative world models [6, 7, 9] and finance agents [19–21] typically *assume* a usable observation stream. We build that missing layer: compile asynchronous evidence into a consumable PIT world bundle; when the view is too thin, force abstention so half-baked context cannot harm decisions. Refusal is the safety valve—not a substitute for seeing, and not the product definition.

Failure modes we target. Without a feed, public models refuse (safe but useless). With a thin price fragment, they often trade and lose (unsafe). With a disclosed but ungated world, they read fields yet disagree on whether to act. The observation layer must therefore (i) make the market *legible* and (ii) attach an enforceable quality valve.

What we build. An anonymized *world-model runtime* emits a JSON world bundle for GPT /DeepSeek /GLM /Gemini-class models. "World model" here means observation compiler + quality-gated runtime (Section 3), not next-state dynamics. The bundle is **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence IDs). WMI/ACWMI and `should_ai_abstain` make thin worlds machine-readable.

How we evaluate. **RQ1 (primary—seeing):** Does the Compiled bundle make ready /missing /tilt answers verifiable against PIT ground truth? **RQ2 (safety valve):** When the world is thin, does a hard flag enforce abstention where soft disclosure does not? Four arms on identical days: Compiled (bundle + hard flag), Ungated (same content, no flag), Raw (`mom5`), Blind (no feed). **RQ3 (secondary):** Are vintaged tilts economically nonempty under a transparent no-LLM rule?

Contributions. (1) Observation-layer semantics: epistemic observations, compilation Π_t, and a clean split between *seeing* (bundle) and *gating* (hard flag). (2) Anonymized PIT runtime: vintage collectors, BandPIT previous-close clock, quality-tagged bundles, frozen four-arm harness. (3) Live validation: grounding ≈0.97 vs. Raw collapse; hard-gate thin-abstain 1.0 vs. Ungated ≈0.68; nonempty tilts under a no-LLM probe. Estimand: legibility of the compiled world, plus valve reliability—not ΔCE vs. a strongest trading agent.

2 Related Work

World models. World models learn compact states and often dynamics [6, 7, 9]. We address the complementary observation-side problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy—not a Dreamer/JEPA simulator.

LLM agents in finance. Return prediction [12], open financial LLMs [18], memory- and tool-augmented agents [10, 19–21], and ICAIF retrieval/debate/judge systems [1, 3, 15, 16] improve how agents fetch, debate, and are scored. They still assume a usable observation stream. We study the prior step: compile a world bundle, measure whether models can *see* it, then gate action when they must not act.

Selective prediction and PIT.. Abstention can be optimal under noise [4, 5]; we attach refusal to world quality via a runtime field, not classifier confidence alone. Macro vintages [2] and crypto microstructure [11, 13] motivate multi-band compilation. Bootstrap discipline for the content probe follows [8, 14, 17].

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Table 1: Module map (systems view).

Module	Role
Collectors / stores	Vintage multi-band history
BandPIT compiler	Previous-close Π_t, WMI/ACWMI
Bundle builder	Complete / honest / auditable JSON
LLM adapter	Frozen prompts; temp. 0
Eval harness	Four-arm sampling and tables

3 Observation Layer

Definition 3.1 (Epistemic observation). $O_{j,t} = (x, \tau, q, g, r)$: value, latest-available time, quality, main-view gate, semantic role. Omitting (τ, q, g, r) yields a feature dump, not a world observation.

Definition 3.2 (World bundle). $W_t = \Pi_t(\mathcal{F}_t^{\mathrm{raw}})$ with scalar quality $q(W_t)$ (WMI/ACWMI); $\mathcal{F}_t^{\mathrm{AI}} = \sigma(W_t)$. Compilation Π_t aligns clocks, applies honesty/missingness gates, and assembles band views into consumer JSON.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $\mathrm{WMI}_t = B_t U_t H_t$ aggregates breadth, stability, and honesty; ACWMI supports regime-conditional gating. The Compiled layer exposes `should_ai_abstain` and `thin_world` as first-class fields.

Seeing vs. gating. The bundle is what the model *sees* (completeness, tilts, evidence IDs). The hard flag is how the layer *blocks* action when $q(W_t)$ is too low. Tilts without gates are still a dump; a gate without a readable bundle is only a switch.

Proposition 3.3 (Compilation ≠ dump). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\mathrm{AI}}$: ungated stale evidence can raise volume while lowering honesty.*

Proposition 3.4 (World-conditional abstention). *If non-abstain actions exceed a world-dependent cost, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [4].*

Remark 3.5 (Not a generative WM). *We do not learn $p(s_{t+1} \mid s_t, a_t)$. We deliver a state compiler with a quality-gated refuse interface.*

4 Runtime

Figure 1 and Table 1 summarize the anonymized prototype (paths omitted for double-blind review). Five stages map raw evidence to a consumable bundle; frozen-prompt consumers call an OpenAI-compatible adapter at temperature 0.

Data flow. Table 2 traces the three-band evaluation archive. *Inputs:* exchange klines/funding/OI; vintaged macro risk indices; aggregate stablecoin circulation—each with observation timestamps. *Processing:* BandPIT keeps the latest observation with timestamp $\leq (t-1)$ 23:59, grades fresh/stale/missing, builds WMI/ACWMI, and derives PIT-safe tilts. *Outputs:* a ~3,990-row PIT panel (10 assets × 399 days) and per-day world bundles.

PIT clock and bundle schema. Decision information for calendar day t is reconstructed at $(t-1)$ 23:59; payoff is same-day close-to-close return. Figure 2: macro and alternative stay ready; exchange readiness drops later and defines the thick/thin contrast. Table 3 lists field groups; Listing 1 shows a Compiled example on an exchange-gap day.

Table 2: Archive bands: fetched vs. emitted in the world bundle.

Band	Fetched	Emitted
exchange	OHLCV, funding, OI	mom5; status/age
macro	equity-vol / dollar indices	macro_tilt; status/age
alternative	stablecoin circulation	alt_tilt; status/age
derived	—	WMI, abstain flag, evidence IDs

Table 3: World-bundle field groups (what the LLM sees).

Group	Fields
Timing	`decision_asof`, previous-close protocol
Completeness	ready/limited/missing counts, band status
Quality	WMI, ACWMI, abstain flag, `thin_world`
Content	macro_tilt, alt_tilt
Audit	`evidence_ids`

Listing 1: Compiled world bundle (compact OOS example).


```
{"date":"2026-05-11","asset":"ADA",
 "decision_asof":"2026-05-10T23:59:00",
 "completeness":{"n_ready":2,"n_missing":1},
 "world_model_index":{"wmi":0.015,
   "should_ai_abstain":true,"thin_world":true},
 "macro_tilt":1.0,"alt_tilt":-1.0,
 "audit":{"evidence_ids":[
   "band:exchange:missing","band:macro:ready"]}}
```


Four arms. **Compiled:** full bundle + hard abstain when the quality flag is true (prompt also forbids acting when the world is judged thin). **Ungated:** same content and numeric WMI; soft "you may abstain." **Raw:** `mom5` only—common thin wiring, not a strongest-agent baseline. **Blind:** no feed. Refuse when WMI < 0.2. On this scarce panel max WMI ≈0.093, every OOS day is thin under the production gate—a valve stress test, not proof the layer never opens.

Frozen prompts (gating only). Both Compiled and Ungated consume the same bundle content; only enforcement differs (Listings 2–3). Raw omits world-quality language. Credentials stay in environment variables; prompts, IS/OOS cut, and seeds are frozen for review.

Listing 2: Compiled (excerpt): hard valve.

```
1. If should_ai_abstain is true OR the world is thin,
   you MUST "abstain".
2. Prefer band content over raw momentum.
3. Rationale must match evidence_ids.
```

Listing 3: Ungated (excerpt): soft judgment.

```
1. Judge whether low WMI / missingness makes action
   unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.
```

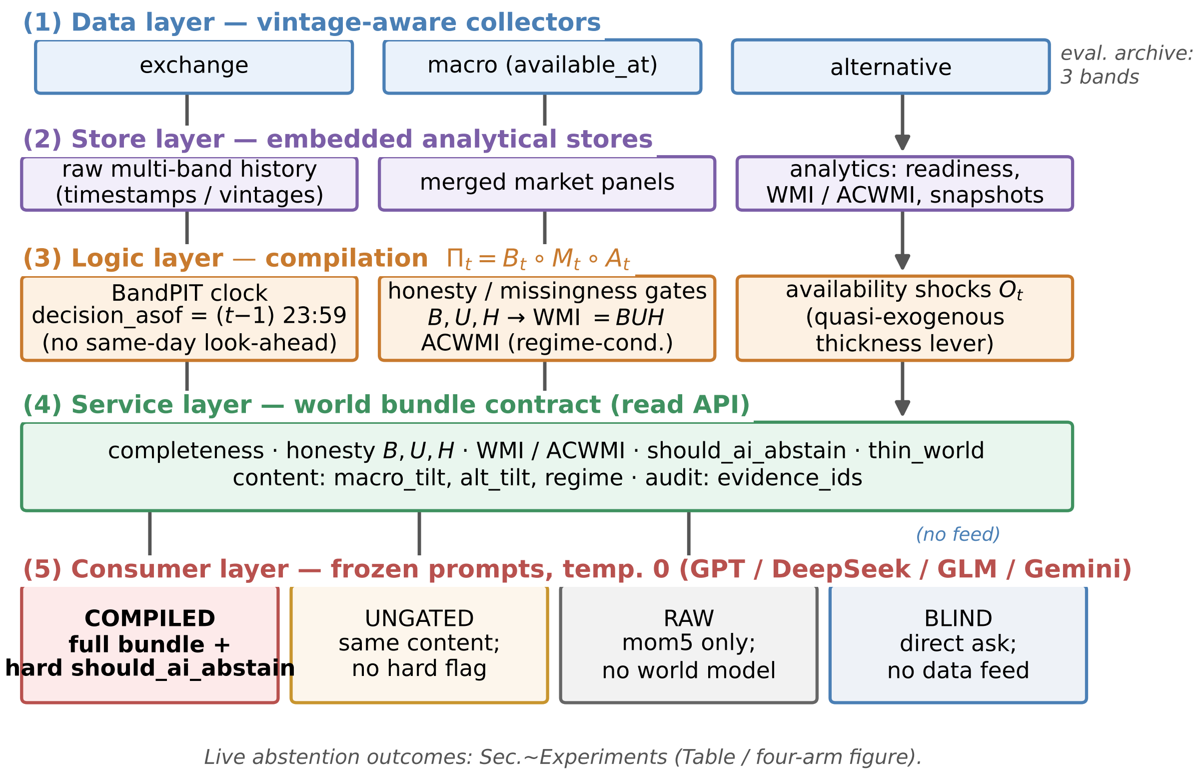

Figure 1: Observation-layer runtime: collectors and stores → BandPIT compilation → world-bundle service → Compiled / Ungated / Raw / Blind consumers.

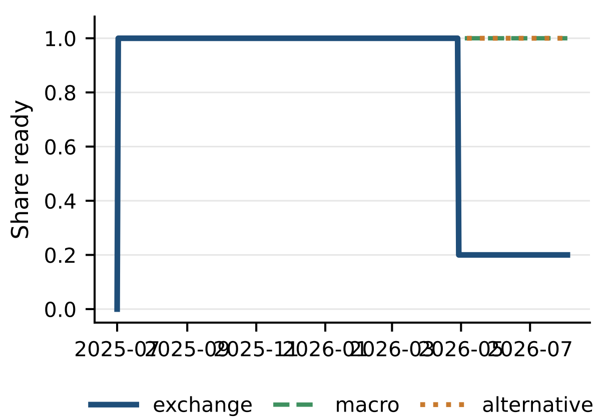

Figure 2: PIT readiness for the three-band archive.

5 Experiments

5.1 Setup

PIT panel ~399 days, 10 liquid names, chronological IS/OOS 200/200 days. Unless noted, live arms share the same stratified 100 OOS asset-days (seed fixed), temperature 0, and four flash/mini vendors. The no-LLM probe uses 199 OOS days after return alignment. Primary metrics for RQ1 are grounding F1/accuracy; for RQ2, thin-world abstain rate (CE as avoided loss, not alpha).

5.2 RQ1 (primary): Seeing—grounding workflow

If the observation layer works, models must report a *checkable* view of the market—not invent coverage from a price fragment, and not confuse "bands present" with "world thick enough to trade."

On 50 OOS asset-days, given Compiled or Raw inputs, each model must (i) list ready bands among exchange/macro/alternative, (ii) list missing bands, (iii) report signs of macro_tilt and alt_tilt, and (iv) judge sufficiency. Steps (i)–(iii) are scored against PIT ground truth.

Table 4: Compiled mean ready F1 0.97, missing F1 0.97, tilt-sign accuracy 0.97; Raw collapses to 0.04, 0.23, 0.19. This is not a trivial copy test in the product sense: Raw has no readiness or tilt fields, so models cannot recover the archive state; Compiled forces the consumer to *use* the disclosed world rather than a momentum guess. Sufficiency chat alone is weak—models may call a 3/3-ready day "enough" while WMI still marks the world thin—so we separate legibility (RQ1) from the hard valve (RQ2).

Table 4: Grounding (50 days): primary seeing result. C = Compiled bundle; R = Raw.

Model	Ready F1		Missing F1		Tilt-sign	
	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Table 5: Thin-world abstention on identical 100 OOS days (Wilson 95% CIs). Blind = 1.0.

Model	Thin C	Thin U [CI]	Abs. R [CI]	ΔCE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	–	+1.07

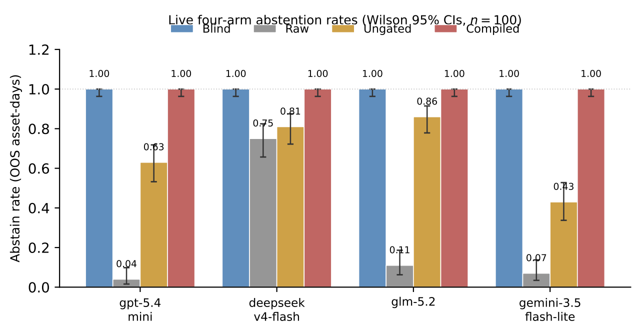

Figure 3: Safety valve: Blind refuses; Raw over-trades; Ungated is unstable; Compiled enforces the hard gate.

5.3 RQ2 (safety valve): Hard refuse when the view is thin

Legibility does not imply safe action. Table 5 and Figure 3: on the same 100 days, Compiled abstains on every thin day (1.0); Ungated soft-judges at mean 0.68 (Gemini 0.43); Raw over-trades (Table 6); Blind abstains 1.0 with no feed. The costly failure mode is the middle case: half-baked Raw context makes models act and lose. Mean ΔCE (Compiled−Raw) +1.07 is avoided loss under the valve, not an alpha claim. Full-OOS (n=2000, all thin) reproduces Compiled 1.0 and Ungated mean ≈0.56.

Scarce-panel note (once). Production Compiled never opens at WMI <0.2 here. On band-thick days (3/3 ready ≡ WMI ≥ 0.05; n=1224), Ungated keeps content without the hard boolean. Four-vendor mean CE ≈+0.29; block-bootstrap mean ≈+0.31 (CIs include zero): a content lower bound, *not* Compiled-open trading—the prompt's thinness clause can still force abstain. Denser vintages can raise the WMI floor without changing layer semantics.

Table 6: Raw action mix (100 days): half-baked context drives directional mass.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Table 7: No-LLM content probe (secondary).

Policy	Sharpe	CE
Buy-and-hold	−1.404	−1.163
Momentum	0.096	−0.206
World-bundle tilts	**0.764**	**+0.130**
Tilts − Momentum	–	0.334 (p=0.034)

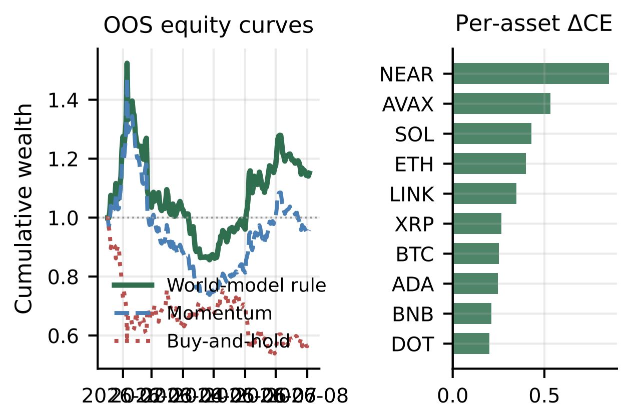

Figure 4: Secondary content probe: compiled tilts beat momentum; ΔCE positive for every asset.

5.4 RQ3 (secondary): Nonempty bundle fields

A transparent no-LLM rule adds vintaged macro_tilt / alt_tilt to return inputs shared with momentum (Table 7, Figure 4). OOS: Sharpe 0.764 / CE +0.130 vs. momentum (0.096, −0.206) and buy-and-hold (−1.404, −1.163); ΔCE=0.334 (p=0.034). LOBO attributes the gap to band content; a long non-vintaged audit shows no edge. This supports nonempty fields inside the observation layer, not an LLM trading ranking.

5.5 Discussion

What carries the paper. (i) *The bundle is legible:* grounding recovers a PIT-checkable market view (RQ1)—the primary product claim. (ii) *Thin views must not drive action:* soft text fails; the hard valve works across vendors (RQ2). (iii) *Fields are nonempty* under a no-LLM probe (RQ3). Relative to FinMem /FinAgent and ICAIF agent stacks [1, 3, 15, 16, 20, 21], we contribute the observation layer those systems assume.

Why grounding is the right primary metric. Trading PnL confounds observation quality with policy skill. Grounding asks a prior question: can the consumer state what is ready, missing, and signed, in a form that can be audited? Raw shows that without the bundle, public LLMs cannot reconstruct that view; Compiled shows they can. The valve then answers the next question: when the disclosed world is thin, will soft judgment stop action? Empirically, no.

Design rationale. (i) Ship a world bundle, not a feature dump. (ii) Previous-close PIT for recomputable `decision_asof`. (iii) Four arms separate seeing+gating, seeing without gating, thin wiring, and no feed.

Threats. Compiled thin abstention is quality-gate obedience by design. The archive is scarce; open-slice Ungated is not Compiled-open; CE bootstrap CIs include zero. Grounding uses 50 days and does not score Ungated separately. Vendors are flash/mini at temperature 0; Raw is `mom5` for the ladder, not dominance over tool agents. Decision/execution stacks can sit on top once denser bands lift WMI.

6 Conclusion

Public LLMs need a market they can see before they decide. We contribute an observation layer that compiles asynchronous crypto evidence into a consumable PIT world bundle, and forces abstention when the view is too thin. Live, grounding recovers ready/missing/tilt at ≈0.97 under Compiled vs. Raw collapse; the hard valve abstains 1.0 on thin days where soft judgment does not. Compile so models can see; refuse when they must not act.

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